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ORIGINAL PAPER

Exploring the mechanism of tartrazine dye adsorption by Parthenium hysterophorus biochar for sustainable textile wastewater treatment

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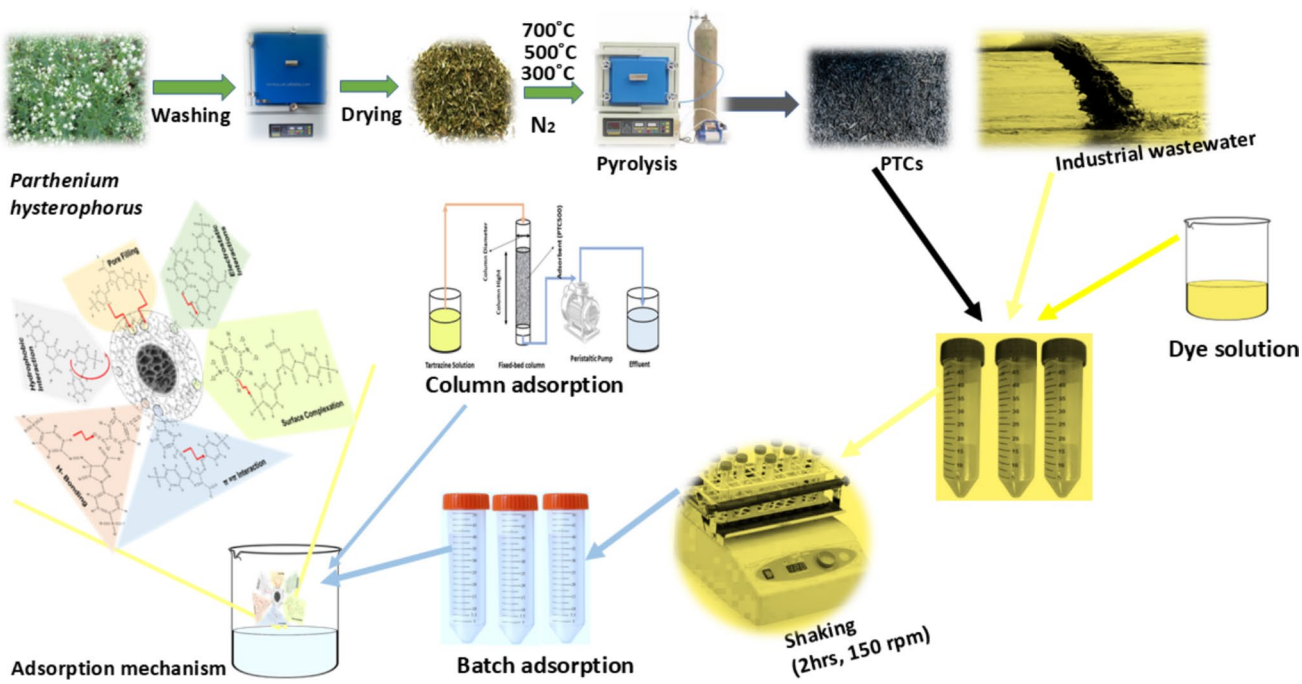
Abstract

In the current study, *parthenium hysterophorus-derived* biochars (PTCs) were used as low-cost and environment-friendly sorbents in batch mode and fixed-bed columns to assess the adsorption behavior of tartrazine dye (TTZ) by varying operational settings and analyzing mathematical models. In the batch adsorption study, the Langmuir and pseudo-first-order models provided the best expression for TTZ adsorption onto PTCs with an uptake capacity of 95.57 mg/g at maximum. In the fixed bed column, the sorption of TTZ was promoted with greater bed height, reduced flow velocity, and higher dye concentration. Comparing the column models (Yoon-Nelson, Thomas, and Clark), the Clark model provided the best fit with the experimental data and effectively represented the adsorption process. The accompanying experimental data agreed with the breakthrough curves. The maximum adsorption capacity achieved with the column process was 8.32 mg/g. In contrast to other adsorbents, PTCs showed excellent adsorption efficiency and a strong potential for reusability.

Graphical abstract



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Keywords *Parthenium hysterophorus* · Dye wastewater treatment · Environmental remediation · Column adsorption modeling · Adsorption kinetics

Introduction

One of the biggest problems in the world is contaminated industrial wastewater, which causes diseases and leads to the deaths of thousands of people every day. The textile industry is a leading contributor to wastewater contamination. Different unit operations and processes adopted in the wet processing industries, like de-sizing, bleaching, dyeing, and printing, use several different chemicals that ultimately end up in the textile wastewater (Nawaz et al. 2013). Types of chemicals could be inorganic, organic, or polymeric, including enzymes, salts, soda, detergents, and color dyes (Pandey et al. 2023).

Today, there are over 12,000 different synthetic dyes available, which are grouped into three types: (i) cationic dyes, including all basic dyes; (ii) anionic dyes, further comprising direct, reactive, and acidic dyes; and (iii) non-ionic dyes, including dispersed dyes (Alegbe and Uthman 2024). Synthetic dyes can be harmful and carcinogenic, and lead to serious health problems like allergies, genetic changes, and inherited diseases. Also, they demonstrate strong physical and chemical properties, heat resistance, and light behavior stabilization (Yasmeen et al. 2023). Tartrazine (TTZ) is a synthetic dye and belongs to the azo dye group of organic compounds with a specific (–N = N–) functional group. Azo dyes are often called ice or magic colors because they need a low temperature to work. These dyes have bright colors because they contain special ring structures (Barciela et al. 2023). Due to their strong azo bond, these dyes do not biodegrade in conventional aerobic biological wastewater treatment plants (WWTP) and need either very long hydraulic retention times or the inclusion of anoxic or anaerobic chambers (Malik et al. 2021).

Biochars are eco-friendly materials that can be easily synthesized through the pyrolysis (burning in the absence of oxygen) of organic materials, such as agricultural and animal waste. Biochars are cost-effective, rich in carbon, and possess various active groups on their surface, exhibiting stable physical and chemical properties. Biochars are effective in removing harmful pollutants, heavy metals, and chemicals such as dyes from wastewater streams. The effectiveness of biochar is related to its physico-chemical features, which are affected by the method of preparation (Bankole et al. 2024). The preparation of biochar at high temperatures through pyrolysis improves its removal efficiency. After the pyrolysis process, surface area change, production of tiny holes, and the appearance of special functional groups are observed in addition to increased charge density, reactivity, and stability (Rambhatla et al. 2025). Even though agricultural weeds are common and can badly harm farming and the economy, they can be used to create biochar materials that can clean polluted water (Song et al. 2024). Parthenium, commonly called congress grass, is well-known to be harmful to animals that eat it and can negatively affect other plants (Jamil et al. 2022). According to recent research, activated carbons and biochars made from various precursors have widely differing adsorption capabilities. The sawdust-derived biochar prepared at 400 °C only produced 0.80 mg/g adsorption, but Psidium guajava seed activated carbon at 600 °C resulted in 1.62 mg/g of adsorption (Almeida et al. 2024). Activated carbons prepared from cassava sievate at 550 °C resulted in 20.83 mg/g of adsorption (Chukwuemeka-Okorie et al. 2021) and Moringa oleifera 500 °C resulted in 2.99 mg/g (Okeola et al. 2017), which is a good performance.

This study fills important research gaps by examining biochar generated at different temperatures (PTCs) from the noxious weed '*Parthenium hysterophorus',* being a novel, affordable, and environmentally acceptable adsorbent for the TTZ dye removal, providing a long-term solution for controlling the biomass of invasive weeds while addressing water pollution. No chemicals were added, and direct carbonization was performed because it does not leave any chemical residue, resulting in a green biochar. This paper offers a thorough analysis of both batch and fixed-bed column systems, bridging the technical gap between laboratory-scale research and real-world applications. To better understand the adsorption capacity and kinetics, different isotherm models are studied and fitted to experimental data. For each PTC, the reuse potential of these agro-weed adsorbents was also investigated. The reusability and affordability of PTCs further highlight their potential as an environmentally friendly industrial wastewater treatment option that supports the circular economy and environmental restoration. Through the integration of waste valorization and efficient dye removal, this study tackles real-world issues in continuous-flow systems and improves the field of biochar-based adsorption.

Materials and methods

Chemicals and composition of TTZ dye solution

The anionic azo-dye TTZ (trisodium (4E)-5-oxo-1-(4-sulfonatophenyl)-4-[(4-sulfonatophenyl) hydrazono]-3-pyrazolecarboxylate), commonly known as the stabilizer E102, was used. It has a molecular weight (534.4 g/mol) and a formula ($C_{16}H_9N_4Na_3O_9S_2$), and was sourced from Sigma Aldrich, UK. Different dye concentrations were prepared in distilled water for initial batch optimization experiments. All other reagent-grade chemicals for adsorbent regeneration, like HCl, H_2SO_4, H_3PO_4, NaCl, NaOH, were also procured from Sigma Aldrich, and were used as such without any processing.

For column and batch studies with real textile wastewater, the sample collection was done from a local textile processing mill, which deals with the fabrication and coloring of polyester strings. The untreated samples were collected and stored in a refrigerator at 4 °C. For sample's characterization, standard methods were used (APHA 2017).

Collection of Parthenium and production of biochar

Parthenium hysterophorus weeds were picked, cleaned to get rid of dirt, and then dried, crushed, and ground into small pieces (< 2 mm). The next step was to slowly carbonize the weed using a furnace (D-550, Vulcan, USA), increasing the temperature at a rate of 5 °C/min until it reached 300 °C, 500 °C, and 700 °C for preparing PTC 300, PTC 500, and PTC 700, respectively. Each temperature was held for 2 h, in oxygen absence, to make three different biochars (Qurat ul et al. 2021).

Batch adsorption study

Experiments were done on an orbital shaker (KJ-201, BD, China) at 150 rpm. The samples were filtered with a Whatman #42 filter paper, followed by a Nylon-cellulose membrane filter (0.22 µm) due to its outstanding purity, uniformity, and low chemical interaction, which guarantees repeatability and minor dye retention (Ebeid et al. 2024). The samples were then tested using a UV–visible spectrophotometer at 470 nm (6300PC, Chincan, China) to determine the remaining concentration of TTZ (Mondal et al. 2017). Control samples (without PTCs) were also tested in each group. Three replicas of each adsorption test were performed to authenticate the findings. The amount of TTZ removed and absorbed was calculated using Eqs. (1) and (2), respectively (Hasan et al. 2024).

$$\text{Removal}(\%) = \frac{C_i - C_e}{C_i} \times 100 \tag{1}$$

$$\text{Adsorption capacity (mg/g)} = \left[\frac{C_i - C_e}{W}\right] \times V \tag{2}$$

where 'C_i' is the initial TTZ concentration (mg/L), 'C_e' is the equilibrium TTZ concentration (mg/L), 'W' is the weight of PTC (g), and 'V' is the volume of solution (L).

Operating parameters to achieve the highest TTZ removal were optimized in a systematic set of experiments. Initially, a pH range of 2–14 was explored by keeping other parameters constant: 10 mg/L TTZ concentration, 1.5 g/L PTC dose, 25 ± 2 °C wastewater temperature, and 24 h adsorption time. Then, the initial concentration of TTZ was varied between 0.5–15 mg/L, keeping other parameters constant, followed by PTC adsorbent dose optimization between 0.2–3.5 g/L. Finally, the contact time between 10–1440 min was evaluated and optimized (Hasan et al. 2024).

Adsorption isotherm and kinetic models

Four different isotherm models (Langmuir, Temkin, Freundlich, and Dubinin-Radushkevich) were applied to TTZ adsorption data to evaluate adsorption capacity onto unique PTCs (Qurat ul et al. 2021). The models' non-linear forms were used as specified under the circumstances in Eqs. (3–6) and the energy of bonding is calculated through Eq. (7).

$$\text{Langmuir}: Q_e = \frac{Q_{max} \times K_L \times C_e}{1 + K_L \times C_e} \tag{3}$$

$$\text{Freundlich}: Q_e = K_F C_e^{1/n} \tag{4}$$

$$\text{Temkin}: Q_e = \frac{RT}{B} \text{Ln}\left(AC_e\right) \tag{5}$$

$$\text{Dubunin} - \text{Radushkevich}: Q_e = Q_D \exp\left[-B_D\left\{RT \ln\left(1 + \frac{1}{C_e}\right)\right\}^2\right] \tag{6}$$

$$E = \frac{1}{\left(2B_D\right)^{0.5}} \tag{7}$$

where 'n' is the adsorption strength constant, 'R' is gas constant (8.314 J/mol/K), 'T' is absolute temperature (K), 'B' is heat of adsorption (kJ/mol), 'A' is binding constant (L/mg), 'Q_D' is the Dubinin's adsorption capacity (mg/g), 'B_D' is the energy constant (mol^2/kJ^2), 'E' is the energy of bonding (kJ/mol), and 'Q_{max}' is Langmuir's maximum adsorption capacity (mg/g).

The following three kinetic models, Eqs. (8–10), were used in their non-linear geometries to help explain the TTZ sorption onto PTCs (Hasan et al. 2024).

$$\text{Pseudo first order}: q_t = q_e\left(1 - e^{-k_1' t}\right) \quad (8)$$

$$\text{Pseudo second order}: q_t = \frac{k_2' q_e^2 t}{1 + k_2' q_e t} \quad (9)$$

$$\text{Elovich}: Q_t = \frac{1}{\beta} Ln(\alpha\beta t + 1) \quad (10)$$

where 't' is the contact period (min), 'α' is the rate of adsorption (mg/g-min), 'β' is sorption constant (g/mg), 'q_e' is the equilibrium TTZ concentration (mg/g), and 'k_1' (1/min) and 'k_2' (g/mg-min) are the rate constants for pseudo first order and pseudo second order, respectively.

Reuse of adsorbent

To test the PTC's reusability and recovery capability, six adsorption cycles were performed using the PTC 500 (selected based on maximum adsorption in batch experiments) with desorbing agents: NaCl (0.1 N), NaOH (0.1 N), H_2SO_4 (0.1 N), H_3PO_4 (0.1 N), and HCl (0.1 N). PTC 500 (0.1 g) was loaded with TTZ (50 mg/L) and mixed with the respective desorbing agents at 150 rpm for 1 h. For effective recovery of PTC 500 after washing, two-step screening using a Whatman #42 filter followed by vacuum filtration with a 0.22 µm nylon-cellulose membrane was performed to guarantee total retention of submicron PTC 500 particles and produce a transparent solution appropriate for UV–visible spectroscopy (Irfan et al. 2023). The resulting TTZ aqueous concentration was measured using Eq. (11).

$$\text{Desorption\%} = \text{Total of TTZ desorbed(mg/g)}/\text{Total of TTZ adsorbed (mg/g)} \times 100 \quad (11)$$

Column adsorption study

A crucial aspect of evaluating the viability of adsorbents in industrial applications is their sorption efficacy in continuous systems. Figure 1 outlines the schematic of the laboratory-scale fixed-bed column setup. The setup included a vitric column with a 5 cm diameter and a height of 120 cm. Glass fleeces were placed at both ends of the column to prevent adsorbent loss in the treated water. A fixed adsorbent concentration was filled in the column. The column was soaked with de-ionized (DI) water to evacuate the trapped gases between the adsorbent particles. The TTZ was pumped from the bottom of the column to elute through the adsorbent with a peristaltic pump (Longer, China). A membrane filter (0.45 µm) was used to remove any remaining traces of the adsorbent from the treated effluent. In the column study, operating parameters for achieving the highest TTZ removal were optimized through a systematic set of experiments. Initially, by changing other parameters, pH was kept constant at 6, TTZ concentration varied between 50–100 mg/L, between 10–20 cm bed height, and the mass of PTC 500 used was between 50–150 g. The flow rate was varied between 8.33 and 12.50 mL/min. The volume of the solution varied between 12,000–18,000 mL. Finally, the contact time, ranging from 10 to 1440 min, was evaluated and optimized.

Breakthrough curve study

Breakthrough curve (50%) determined the buckling behavior of TTZ onto PTC 500 (selected based on maximum adsorption in batch experiments) in column, defining (C_t/C_o Vs

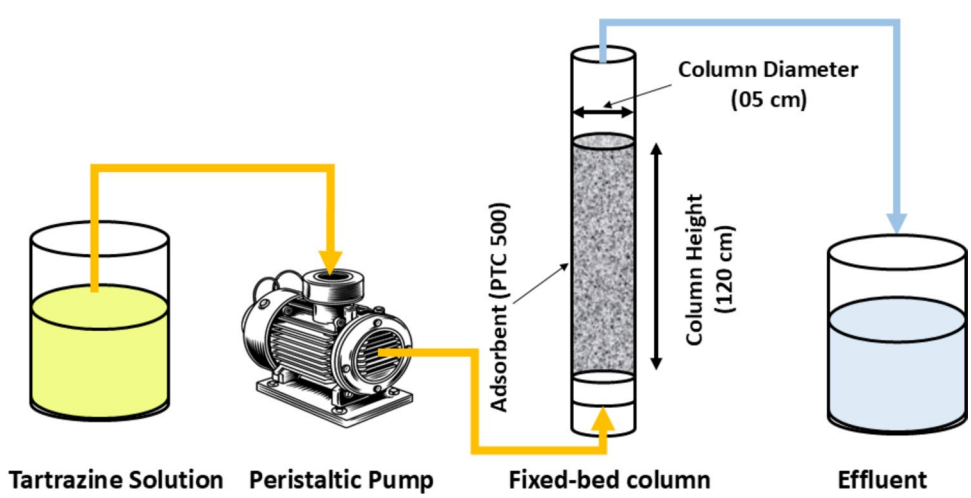


Fig. 1 Schematic of TTZ dye adsorption column setup

T) (Almeida et al. 2024). TTZ mass adsorbed (mg/g) was calculated using the area (A) underneath the breakthrough curve and flow volume (mL) by Eqs. (12–14) (Sadaf and Bhatti 2014).

$$\text{Total mass adsorbed} : q(0.5) = Q \int_{t=0}^{t=0.5} \left(1 - \frac{C_t}{C_o}\right) d_t \quad (12)$$

$$\text{Adsorption capacity} : q_{ac} = {q_{0.5}}/{W} \quad (13)$$

$$\text{Effluent volume} : V = Qt \quad (14)$$

where 'W' indicates the mass of PTC (g) filled in the column. At a pre-determined flow rate, pH, and bed height, a fixed measured quantity of PTC was placed into the column, and an appropriate volume of TTZ solution was injected from the bottom. An automated collector gathered the wastewater, while a ball valve with a flowmeter monitored and managed the flow rate. The removal efficiencies, adsorption capacity, and kinetics were quantified using the numerical nonlinear models of Clark, Thomas, Yoon-Nelson, and Bohart-Adams (Al-Mahbashi et al. 2022) as described in Eqs. (15–18).

$$\text{Clark Model} : {C_t}/{C_O} = \left(1 + (n/A)\exp\left(-K_C \times t\right)\right)^{-1/n} \quad (15)$$

$$\text{Thomas model} : {C_t}/{C_O} = 1/(1 + \exp(K_{TH} \times (q_o \times M - Q_t))) \quad (16)$$

$$\text{Yoon} - \text{Nelson model}: {C_t}/{C_O} = 1/(1 + \exp(K_{YN} \times (t - \tau))) \quad (17)$$

$$\text{Bohart} - \text{Adams model} : {C_t}/{C_O} = 1/(1 + \exp(\frac{K_{BA} Z q_O}{Q} - K_{BA} C_O t)) \quad (18)$$

where 'C_t' is effluent concentration (mg/L), 'C_0' is initial concentration (mg/L), 'K' is rate constant, 't' is time (min), 'A' is adsorption capacity constant and flow rate, 'n' is sorption intensity, 'q_0' is adsorption capacity (mg/g), 'M' is adsorbent mass (g), 'Q' is flow rate (mL/min), and 'τ' is time for 50% breakthrough (min) 'Z' is bed depth (cm).

Results and discussion

Characterization of biochars

Surface morphology examination of PTCs was conducted using a scanning electron microscope (SEM) (Mira 3, Tescan, Czech Republic) with energy dispersive x-ray (EDX) spectroscopy (JED-2300, Jeol, Japan), and the images are shown in Fig. 2. It was observed that the PTCs had exceedingly sporadic surfaces described by a significant level of

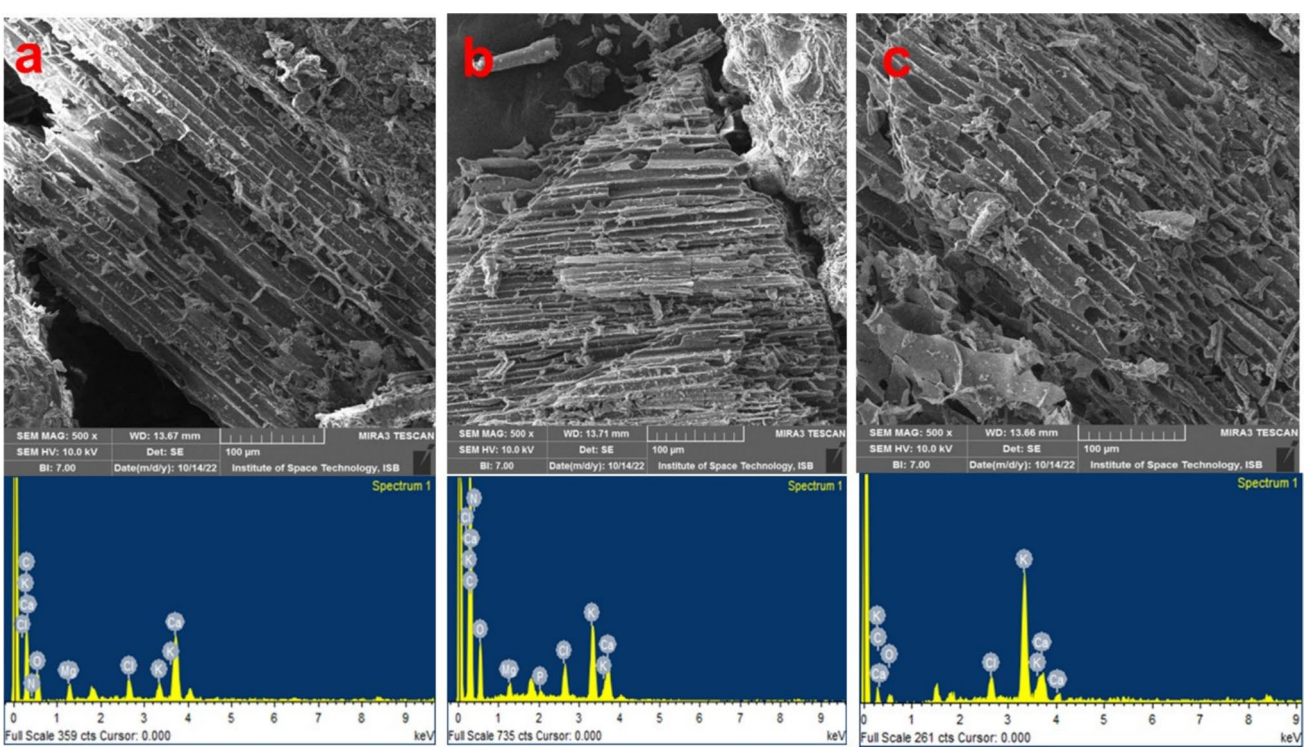


Fig. 2 SEM images and EDX spectra of **a** PTC 300, **b** PTC 500, and **c** PTC 700

heterogeneity, and the appearance of cavities, gaps, breaks, and voids. These qualities of PTCs play a pivotal role in improving their sorption capacity. Moreover, high temperatures during the carbonization and calcination steps caused the discharge of unstable substances from the particles' microstructure. This caused the voids generation on PTC's surfaces, eventually changing to pores that entrap toxic chemicals (Rawat et al. 2021).

The FTIR spectra described different categories of surface functional groups on PTCs, as shown in Fig. 3a. Strong signals at 3306 cm^{-1} and 2980 cm^{-1} show the presence of an alcohol group (Mondal et al. 2017). A peak was found at 1564 cm^{-1}, showing amine's presence in different types of aromatic compounds at 1397 cm^{-1}, 1252 cm^{-1}, and 1106 cm^{-1}. Also, at 876 cm^{-1} and 786 cm^{-1}, respectively, the presence of alkene and halo compounds is noticed (Shams et al. 2020). Though the FTIR spectra for all three PTCs were similar, noticeable aromatic compounds were found in PTC 500 and PTC 700, showing a strong aromatic quality compared to PTC 300. After the adsorption test, peaks shifted from 3306 cm^{-1} to 3670 cm^{-1}, 2980 cm^{-1} to 2936 cm^{-1}, and 1564 cm^{-1} to 1576 cm^{-1} due to the TTZ adsorption on PTCs (Fig. 3b). There are some peaks on 2160 cm^{-1} and 1730 cm^{-1}, showing a bond formation between TTZ and PTCs. The peak at 1576 cm^{-1} shows cyclic alkene, and 1405 cm^{-1} and 1026 cm^{-1} show sulfate and amine binding, respectively (Albadarin et al. 2017).

Adsorption parameters optimization

Adsorption parameters, such as the dose of adsorbent, pH of the solution, adsorbate concentration, and contact duration, are the primary determining factors of the adsorption process. Figure 4 (a-h) illustrates how these factors affect the adsorption capabilities of PTCs. The percentage removal of TTZ increased from 28.7 to 89.7%, 48.4 to 98.9% and 40.8 to 95.4%, while adsorption capacity decreased from 14.3 to 2.8 mg/g, 22.7 to 3.1 mg/g, and 20.4 to 2.8 mg/g for PTC 300, PTC 500, and PTC 700, respectively, with increasing PTCs dose from 0.2 to 3 g/L (Fig. 4 a, b). The higher adsorbent dose guarantees the availability of enough active sites for the adsorbate, resulting in an improvement in removal performance (Ahmed et al. 2024). The adsorption capacity reduction with increasing dose may be attributed to the phenomenon of the adsorbent's mass aggregation. The adsorbent particles can cluster together, reducing the overall surface area available for adsorption and blocking the active sites on the biochar, making them inaccessible to the adsorbate molecules (Ambaye et al. 2020). Based on the removal results, the optimized adsorbent dose is 1.5 g/L.

For pH optimization experiments, the trend was similar for both the adsorption capacity and removal efficiency, as shown in Fig. 4 (c, d). Both peaked at pH 6 for all three PTCs, then stayed constant until pH 8, and subsequently declined till pH 12. The exterior of anionic TTZ is positively charged at low pH levels up to 6, which promotes its adsorption. As the pH rises, the PTC exterior turns negatively charged, and there are more negative sites. Above 8 pH, there was a decrease in adsorption capacity because of the repellent effects of anionic TTZ dye and the negative charges on the PTC's surface (Pirarath et al. 2023). Based on the results, the selected optimized solution pH was 6.

Figure 4 (e, f) illustrates how the adsorption capacities of three distinct PTCs were impacted by the initial TTZ concentration. Percent removal of TTZ remained consistent at around 100% between 0.5–5 mg/L for all PTCs, but then it demonstrated a gradual reduction to 60–85% at 15 mg/L of initial TTZ concentration. A decline in percentage removal occurs when the original dye concentration rises because of the occupation of available active sites. With higher initial concentrations of TTZ, the adsorbent may find it more difficult to extract the dye from the solution due to a increased mass transfer resistance from the higher TTZ concentration (Bakhta et al. 2024). The same pattern was also observed

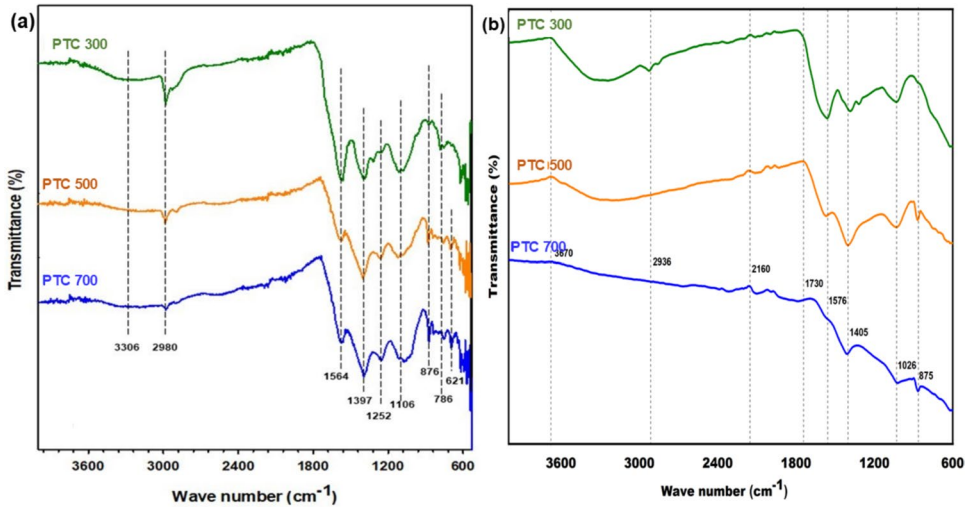


Fig. 3 FTIR spectra of biochars (PTC 300, PTC 500, and PTC 700), **a** before adsorption and **b** after adsorption

in another study where methylene blue dye was removed using biochar made from the leaves and stems of *Lantana camara* L. (Kundu et al. 2024). Interestingly, the adsorption capacity per unit mass often increases with higher initial dye concentrations. This happens because the mass transfer driving force from solution to adsorbent surface becomes stronger, overcoming resistance to adsorption (Yagub et al. 2014). A similar relationship was observed in this study, where adsorption capacity increased from 0.5 to 12 mg/g with increasing initial TTZ concentration. Based on removal efficiency, the selected optimized initial TTZ concentration is 5 mg/L.

The relationship between contact time, TTZ removal and the adsorption capacity by different PTCs is described in Fig. 4 (g, h). Both the removal and adsorption capacity increased between 10–45 min, followed by a plateau up to 1440 min (1 day). Contact time determines the process by which the dye diffuses through the boundary layer, enters the adsorbent's internal pores, and migrates from the main body of the solution to the adsorbent's surface. Longer contact time provides dye molecules a greater opportunity to interact with the adsorbent, which raises the likelihood of adsorption (Amaku et al. 2024). Directed by the saturation of active sites on the adsorbent surface, the removal efficiency becomes independent of contact time. Gradually, an equilibrium is established in the system when the rate of adsorption and the desorption become equal due to fewer available active sites. Hence, the selected optimized contact time was 45 min.

Isotherm modelling

Based on the correlation coefficients (R^2) values (Table 1, Fig. 5 a-d), PTC 500 adsorption information was well-fitted in Langmuir and Freundlich but not in Temkin and D-R applied isotherm models. Langmuir anticipated the highest adsorption capacity (Q_{max}) of 95.57 mg/g for PTC 500, which was more noteworthy than those of PTC 300 (12.16 mg/g) and PTC 700 (59.41 mg/g). The 1/n values anticipated by the Freundlich show for all PTCs were > 1, demonstrating the favorable adsorption of TTZ. Within the Temkin model, the B values were < 8 kJ mol^{-1} for all PTCs, demonstrating that the adsorption was physisorption (Ravindiran et al. 2023). Similar to the Langmuir demonstration, the Q_D values anticipated by the Dubinin-Radushkevich model were more noteworthy for PTC 500 (13.77 mg/g) than for PTC 700PTC 300 (6.57 mg/g) and PTC 300PTC 700 (12.96 mg/g).

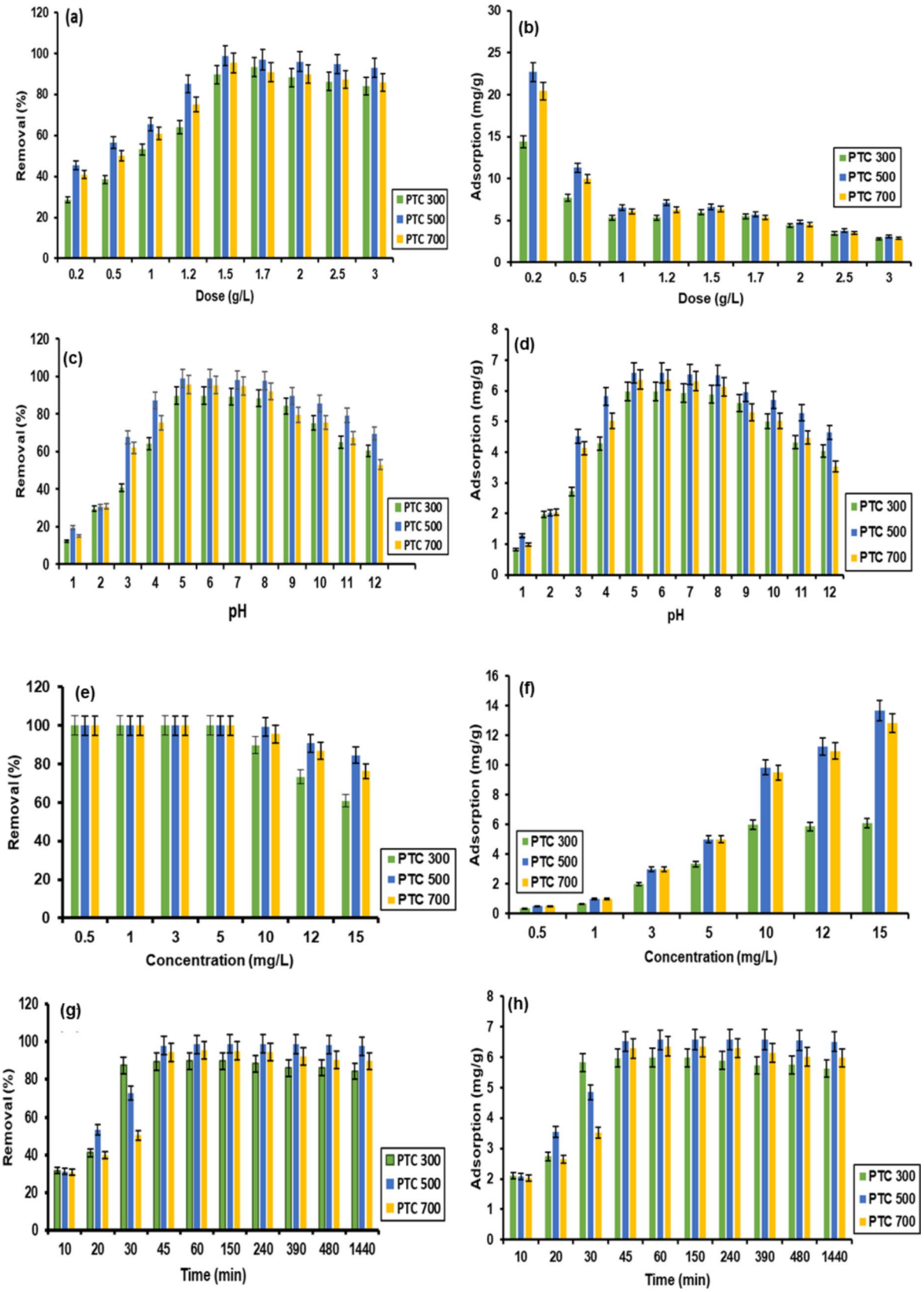
The overall results of the isotherm models demonstrated that PTC 500 showed the greatest adsorption of TTZ as compared to the PTC 300 and PTC 700. The Langmuir isotherm's high R^2 and Q_{max} values reflect a homogeneous monolayer adsorption process and the surface's active spots reflect several exceptional benefits for adsorption (Song et al. 2024). Additionally, the Temkin isotherm model highlights the uniform distribution of binding energies within the PTCs (Banerjee and Chattopadhyaya 2017). On the other hand, the Freundlich, Temkin, and Dubinin-Radushkevich models indicate heterogeneous adsorption, suggesting overall diversity of the PTCs' surfaces. The adsorption of TTZ on PTCs appears to involve multiple mechanisms, as evidenced by the well-fitted isothermal models (Mahmoud et al. 2020), including pore-filling diffusion (Dubinin-Radushkevich model), multilayer adsorption (Freundlich model), and monolayer adsorption (Langmuir model).

The adsorption of TTZ was examined using four kinetic models, intra-particle diffusion (IPD), pseudo-second order (PSO), Elovich, and pseudo-first order (PFO). Figure 5 (e–h) displays the curve fittings for each model, and Table 1 lists the associated parameters. According to the R^2 values (ranging from 0.789 to 0.978), PFO, PSO, and Elovich models fitted well to the TTZ kinetic adsorption data. However, results did not match the intraparticle diffusion model ($R^2 < 0.5$). In comparison to PTC 300 and PTC 700, PTC 500 exhibited higher estimated equilibrium adsorption capacity (q_e) values for both PFO and PSO (6.67 and 7.05 mg/g). Furthermore, PTC 500 exhibited a greater adsorption rate (95.57 mg/g-min). PTC 500 has the maximum adsorption capacity for TTZ when compared to PTC 300 and PTC 700, according to these kinetic data, which are consistent with the isothermal models. This implies the participation of several different adsorption processes. The kinetic data's strong match to the PFO and PSO models suggests that TTZ is adsorbed on all PTCs, but especially on PTC 500, by a combination of physisorption and chemisorption processes. When the highest level of adsorption measured in the current study is equated to other biochars used for the same dye removal from aqueous streams, the current research is superior in performance (Table 2).

Adsorbent recycling and testing with real textile wastewater

In this stage of the study, the optimized operational conditions from Sect. "Adsorption parameters optimization" were selected to remove TTZ from the real textile industrial wastewater through a batch adsorption experiment. The results showed 98% dye removal (Fig. 6b).

In addition, other conventional wastewater pollutants were also significantly removed, as shown in Table 3. The physical structure of PTC 500 entrapped the total suspended solids (TSS) and resulted in a 75% reduction. Similarly, it removed organic contaminants and total dissolved solids (TDS) through adsorption and an ion exchange mechanism. However, further treatment of textile wastewater is required to meet the NEQS permissible limits for biological oxygen


(a)
Removal (%)
Dose (g/L)
0.2 0.5 1 1.2 1.5 1.7 2 2.5 3
PTC 300
PTC 500
PTC 700
(b)
Adsorption (mg/g)
Dose (g/L)
0.2 0.5 1 1.2 1.5 1.7 2 2.5 3
PTC 300
PTC 500
PTC 700
(c)
Removal (%)
pH
1 2 3 4 5 6 7 8 9 10 11 12
PTC 300
PTC 500
PTC 700
(d)
Adsorption (mg/g)
pH
1 2 3 4 5 6 7 8 9 10 11 12
PTC 300
PTC 500
PTC 700
(e)
Removal (%)
Concentration (mg/L)
0.5 1 3 5 10 12 15
PTC 300
PTC 500
PTC 700
(f)
Adsorption (mg/g)
Concentration (mg/L)
0.5 1 3 5 10 12 15
PTC 300
PTC 500
PTC 700
(g)
Removal (%)
Time (min)
10 20 30 45 60 150 240 390 480 1440
PTC 300
PTC 500
PTC 700
(h)
Adsorption (mg/g)
Time (min)
10 20 30 45 60 150 240 390 480 1440
PTC 300
PTC 500
PTC 700

◂**Fig. 4** Effects of **a**, **b** PTC dose (pH = 6, TTZ = 10 mg/L, time = 1440 min, temp = 25 °C) **c**, **d** pH (TTZ = 10 mg/L, PTC = 1.5 g/L, time = 1440 min, temp = 25 °C), **e**, **f** TTZ initial concentration, (PTC = 1.5 g/L, pH = 6, time = 1440 min, temp = 25 °C) and **g**, **h** contact time (TTZ = 10 mg/L, pH = 6, PTC = 1.5 g/L, temp = 25 °C) on removal and adsorption of TTZ by PTC biochars (PTC 300, PTC 500, PTC 700)

demand (BOD) and chemical oxygen demand (COD) parameters. Six sequential adsorption–desorption cycles were used to comprehensively assess the regeneration and reusability of the produced PTC 500 utilizing five distinct 0.1 N regenerative agents (NaOH, NaCl, H_2SO_4, H_3PO_4, and HCl). Before being used again, the dye-loaded PTC 500 was cleaned with DI water, processed with the appropriate reagent following each sorption cycle, and oven-dried at 80 °C.

Figure 6a illustrates how the regeneration effectiveness changed based on the reagent used. The PTC's ability to adsorb after NaOH cleaning was 93.84 mg g^{-1} (98%) in the first cycle and progressively dropped to 93.82 mg g^{-1} (97.9%) after six cycles, which equates to a regeneration efficiency of almost 98%. NaCl also produced a somewhat reduced regeneration efficiency, retaining 75% after six cycles. However, the regeneration of H_2SO_4 and H_3PO_4 demonstrated a considerable persistence of adsorption capacity 60% and 30% respectively, most likely as a result of significant structural changes and the depletion of surface functional groups (Irfan et al. 2023). On the other hand, the most noticeable reduction in reusability was caused by HCl treatment; after six cycles, capacity decreased from 19.15 to 19.06 mg g^{-1}, or barely 20% efficiency. According to the comparable findings, the best regenerants for preserving PTC 500 reusability are NaOH and NaCl. The modest decrease in all reagents can be explained by partial surface oxidation during serial regeneration, pore obstruction, or decreased exposure of adsorption sites (Khalid and Inam 2024). These findings alleviate the environmental concerns regarding the safe disposal of the used adsorbent, as PTC 500 can repeatedly be reused for TTZ adsorption from several batches of textile wastewater.

Column study

In fixed-bed columns, an adsorbent is placed inside a column, and under carefully regulated circumstances, a dye solution with a known influent concentration is added at a steady flow rate. The effluent concentration is then periodically checked as a function of time. Initially, the adsorbent successfully retains the dye molecules, resulting in a very low effluent concentration. However, as time passes, the adsorption sites close to the column entrance gradually get saturated, enabling the dye to be present in the effluent. The effluent concentration steadily rises as saturation spreads throughout the bed, finally approaching the influent

Table 1 Parameters for TTZ adsorption onto biochars (PTC 300, PTC 500, PTC 700) determined using isothermal and kinetic models

Isothermal models

	Langmuir			Freundlich			Temkin			Dubinin Radushkevich		
	R^2	Q_{max}(mg/g)	K_L(l/mg)	R^2	1/n	K_F(mg/g)	R^2	B(kJ/mol)	A(l/mg)	R^2	Q_D(mg/g)	ε(J/mol)
PTC 300	0.975	12.16	0.07	0.952	1.44	1.03	0.921	1.89	1.60	0.969	6.57	1
PTC 500	0.999	95.57	0.01	0.998	1.07	1.12	0.855	3.78	1.36	0.958	13.77	1
PTC 700	0.999	59.41	0.01	0.997	1.12	1.19	0.870	3.60	1.39	0.955	12.96	1

Kinetic models

	Pseudo-first-order			Pseudo-second-order			Elovich			IPD		
	R^2	q_e(mg/g)	k_1(min^{-1})	R^2	(mg/g)	k_2(g/mg-min)	R^2	α(mg/g-min)	β(g/mg)	R^2	C(mg/g)	k_I(mg/g-$min^{-0.5}$)
PTC 300	0.916	5.59	0.05	0.845	6.24	0.014	0.701	45.05	1.74	0.20	3.46	0.100
PTC 500	0.978	6.67	0.04	0.925	7.05	0.009	0.789	10.52	1.28	0.31	3.54	0.120
PTC 700	0.933	6.29	0.03	0.872	6.67	0.003	0.734	5.08	1.26	0.30	3.16	0.124

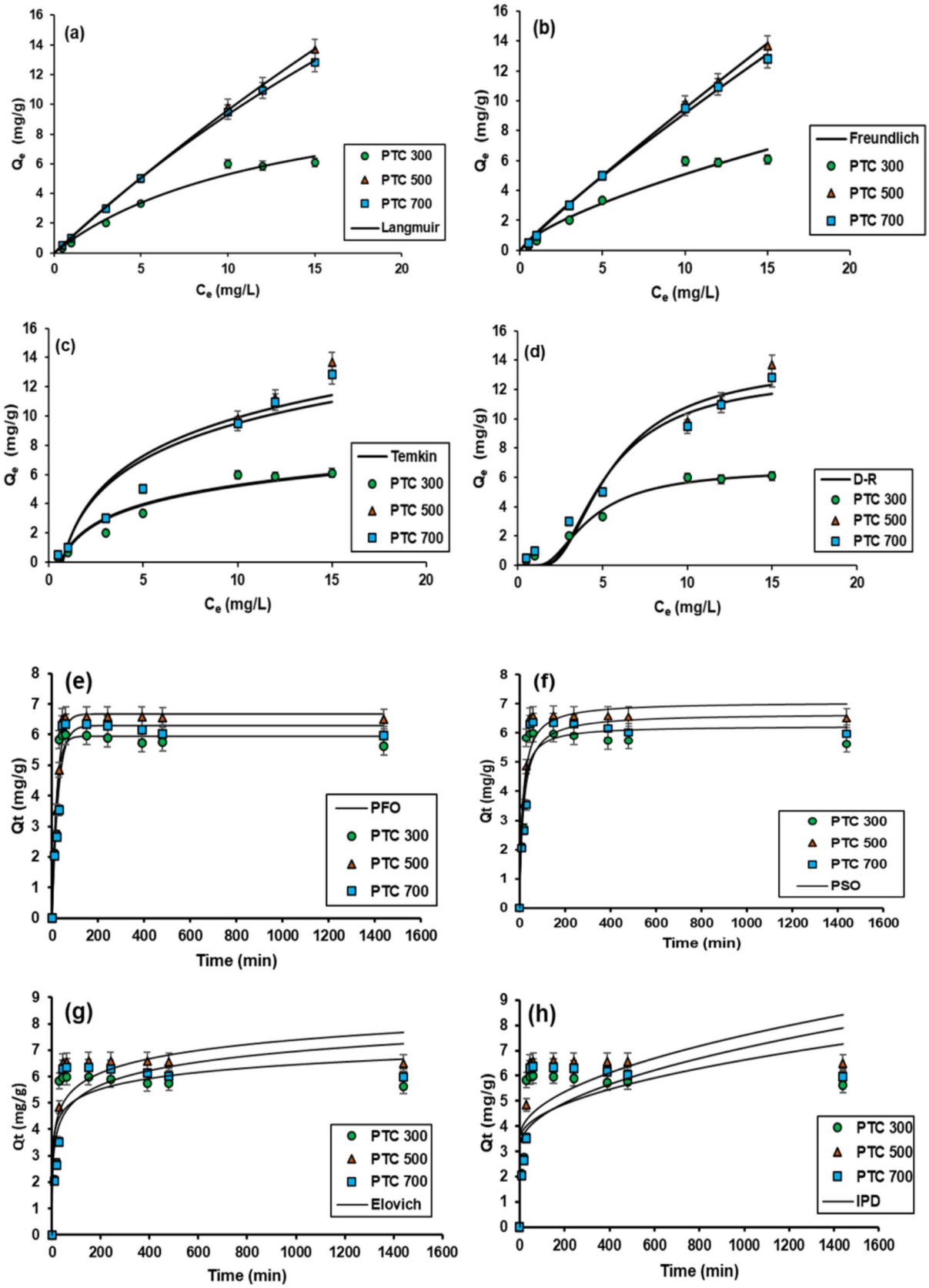


Fig. 5 Non-linear fittings of isothermal models, **a** Langmuir, **b** Freundlich, **c** Temkin, **d** Dubinin Radushkevich (D-R), and kinetic models, **e** pseudo-first-order, **f** pseudo-second-order, **g** Elovich model, and **h** intra-particle diffusion, on TTZ adsorption onto PTC 300, PTC 500, PTC 700

Table 2 Comparison of adsorption capacity of biochar (PTC 500) for the removal of TTZ used in this study with previous studies

Adsorbent	Pyrolysis temperature (°C)	Adsorption capacity (mg/g)	Reference
Commercial biochar	–	4.17	(Soran et al. 2023)
Sawdust from *Pinus halepensis* wood	400	0.80	(Rzig et al. 2024)
Activated carbon from *Cassava sievate*	550	20.83	(Chukwuemeka-Okorie et al. 2021)
Activated carbon from guava (*Psidium guajava*) seeds	600	1.62	(Almeida et al. 2024)
Activated carbon from *Moringa oleifera*	500	2.99	(Okeola et al. 2017)
Parthenium hysterophorus biochar	500	95.57	This study

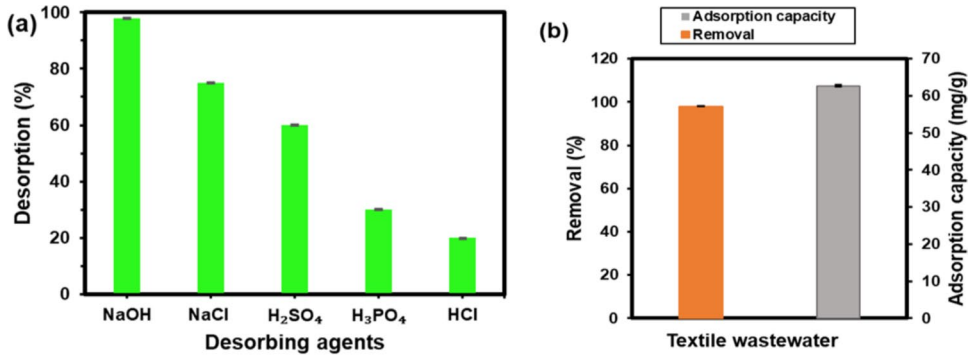


Fig. 6 **a** Desorption of TTZ by different desorbing agents, and **b** removal and adsorption of TTZ from textile wastewater by biochar (PTC 500)

Table 3 Real textile wastewater characteristics before and after adsorption through PTC 500

Parameters	NEQS	Inlet real textile wastewater	After adsorption	% Removal
COD (mg/L)	400	4,120 ± 2	1,030 ± 2	75%
BOD (mg/L)	250	1,550 ± 2	525 ± 2	65%
TDS (mg/L)	3500	4,100 ± 1	2,870 ± 1	30%
pH	6–9	9.5 ± 0.1	8 ± 0.1	–
TSS (mg/L)	400	612 ± 0.1	153 ± 0.1	75%
TTZ (mg/L)	N/A*	0.95 ± 0.01	0.026 ± 0.01	98%

**N/A* Not available (0.1 mg/L as per WHO), *NEQS* National environmental quality standards

concentration and signifying column collapse. The typical breakthrough curve, which is depicted in Fig. 7, is produced by plotting the normalized effluent concentration (C/C_0) against time. This curve depicts the effectual mass transfer activity and functionality of the adsorption column (Ali et al. 2022a).

In the column experiment, we took the PTC 500 as the best adsorbent from the batch study results. When effluent concentration (C_t) reached 50% of influent ($C_t/C_0 = 0.5$), that point was marked as breakthrough ($t_{0.5}$) for this study (Almeida-Naranjo et al. 2024). Different operating conditions were studied as described in Table 4.

Experiments were carried out at pH 6 with PTC 500, a column bed height of 10–20 cm, and vent concentration of 50–100 mg/L to investigate flow rate effects on TTZ column adsorption (Fig. 7a). The breakthrough time dropped from 47.6 to 18.8 min, with increasing flow rate from 8.33 to 12.50 mL/min. However, adsorption capacity declined due to a shorter adsorbent and adsorbate interaction time (Yan et al. 2018).Adsorbent packing density plays a significant role in dye removal. Several bed heights were used in the experiments: 10 cm (50 g), 15 cm (100 g), and 20 cm (150 g) under consistent flow rate (8.33 mL/min) and initial dye concentration (100 mg/L) (Fig. 7b). With increasing bed height from 10 to 20 cm, adsorption capacity increased from 7.94 to 8.32 mg/g and the breakthrough time amplified from 47.6 to 149.8 min. Increasing bed height provides more adsorption sites; however, due to the uneven distribution of the TTZ solution, adsorption capacity was reduced, and a shorter bed height of 10 cm resulted in faster saturation. Higher bed heights produced a breakthrough curve that was flatter and rose more slowly, indicating a larger mass transfer zone and improved adsorption. The ideal bed height for maximal adsorption was determined to be 20 cm, and this setting was selected for additional testing. Similar adsorption trends with changing bed heights were observed in another study (Fernandez et al. 2023).

Impacts of initial dye concentration (50–100 mg/L) on breakthrough curve were investigated using a fixed

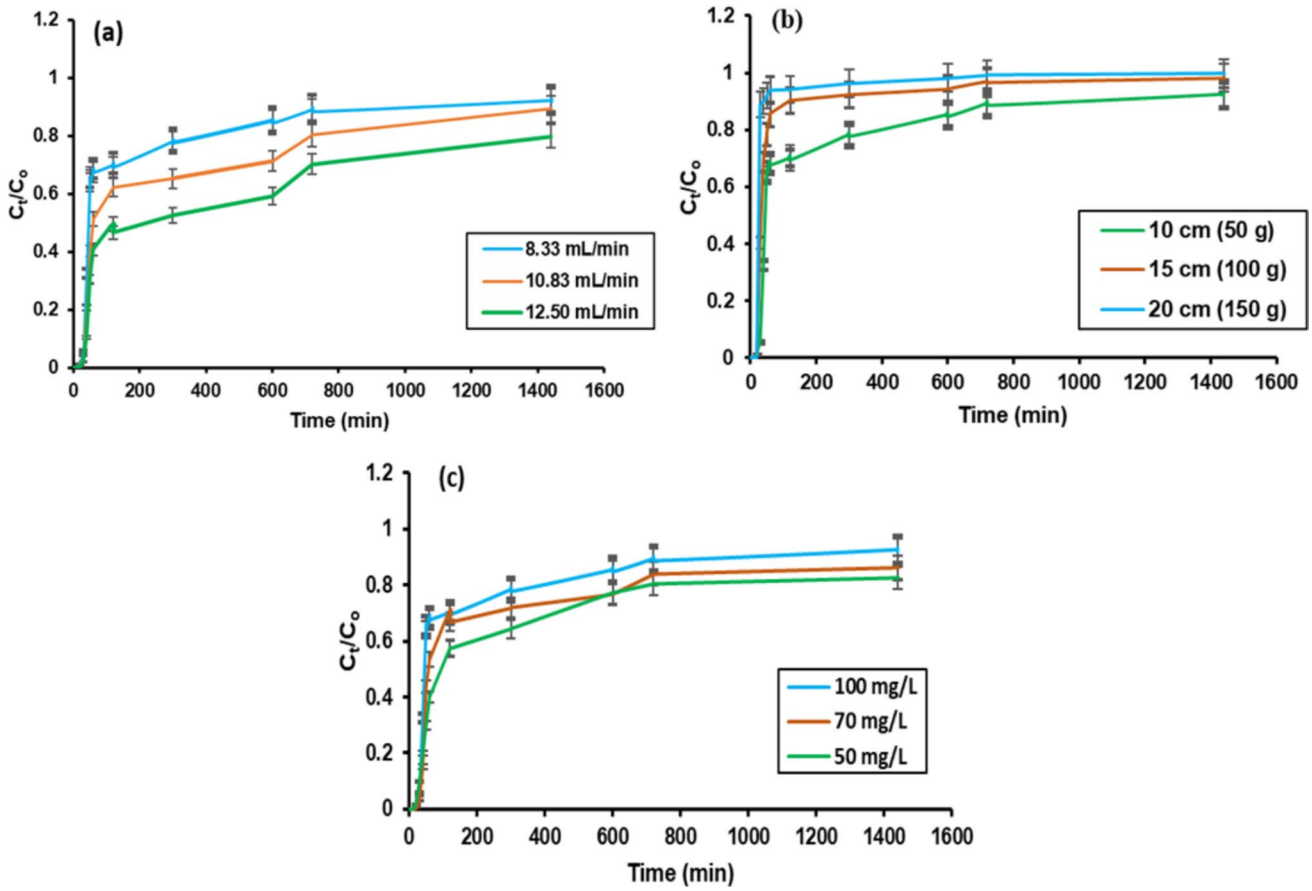


Fig. 7 Effects of **a** flow rate, (pH = 6, time 1440 min, temp = 25 °C, Bed height (mass of adsorbent) = 10 cm (50 g), 15 cm (100 g), 20 cm (150 g), initial concentration of TTZ = 100 mg/L, 70 mg/L,50 mg/L), **b** bed height (mass of adsorbent) (pH = 6, time 1440 min, temp = 25 °C, flow rate = 8.33 mL/min, 10.83 mL/min, 12.50 mL/min, initial concentration of TTZ = 100 mg/L, 70 mg/L, 50 mg/L), and **c** the initial concentration of TTZ (pH = 6, time 1440 min, temp = 25 °C, Bed height (mass of adsorbent) = 10 cm (50 g),15 c m(100 g),20 cm(150 g), flow rate = 8.33 mL/min, 10.83 mL/min, 12.50 mL/min, on adsorption by biochar (PTC 500)

Table 4 Column parameters with different bed heights, flow rates, and vent concentration

Vent conc. (mg/L)	Flow rate (mL/min)	Bed height (cm)	Mass of PTC 500 (g)	Breakthrough curve point (50%) (min)	Treated volume (mL)	Adsorption capacity (mg/g)
50	8.33	10	50	15.6	11,999	2.61
70	8.33	10	50	18.9	11,999	3.15
100	8.33	10	50	47.6	11,999	7.94
100	8.33	10	50	47.6	11,999	7.94
100	10.83	10	50	30.6	15,595	6.63
100	12.50	10	50	18.8	18,000	4.70
100	8.33	10	50	47.6	11,999	7.94
100	8.33	15	100	99.2	11,999	8.27
100	8.33	20	150	149.8	11,999	8.32

bed height of 20 cm and stream flow rate of 8.33 mL/min (Fig. 7c). With high initial concentration (100 mg/L), adsorption driving force become high, resulting in quicker adsorption (Singh et al. 2024). The breakthrough was achieved in a shorter time of only 15.6 min, and adsorption capacity improved from 2.61 to 7.94 mg/g.

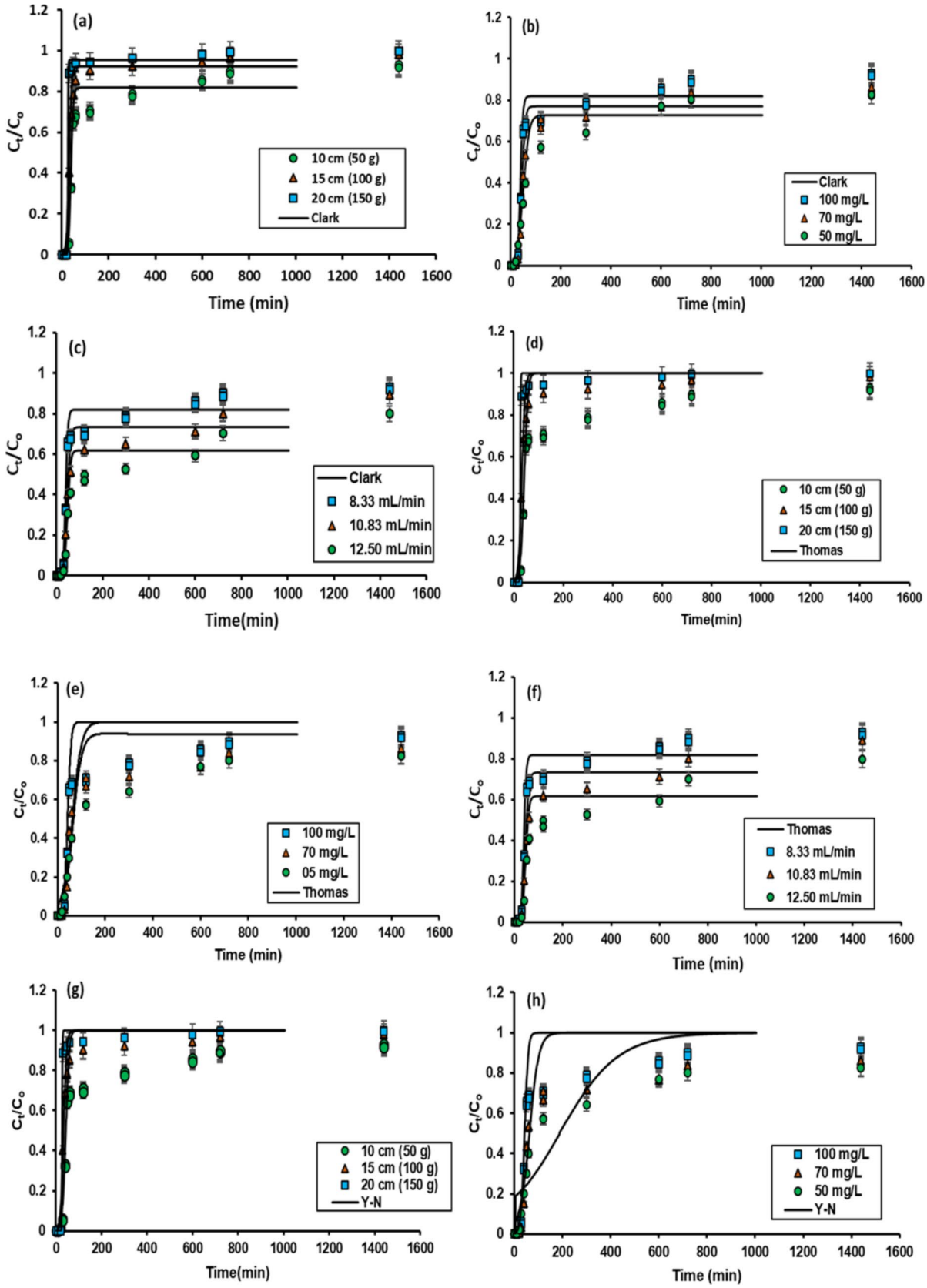


Fig. 8 Non-linear fittings of breakthrough curves models, (pH = 6, time 1440 min, temp = 25 °C, Flow rate = 8.33 mL/min, 10.83 mL/min, 12.50 mL/min, initial concentration of TTZ = 100 mg/L, 70 mg/L, 50 mg/L, bed height (mass of adsorbent) = 10 cm (50 g),15 cm (100 g), 20 cm (150 g)) **a–c** Clark, **d–f** Thomas, **g–i** Yoon–Nelson, **j–l** Bohart-Adams on TTZ adsorption onto biochar (PTC 500)

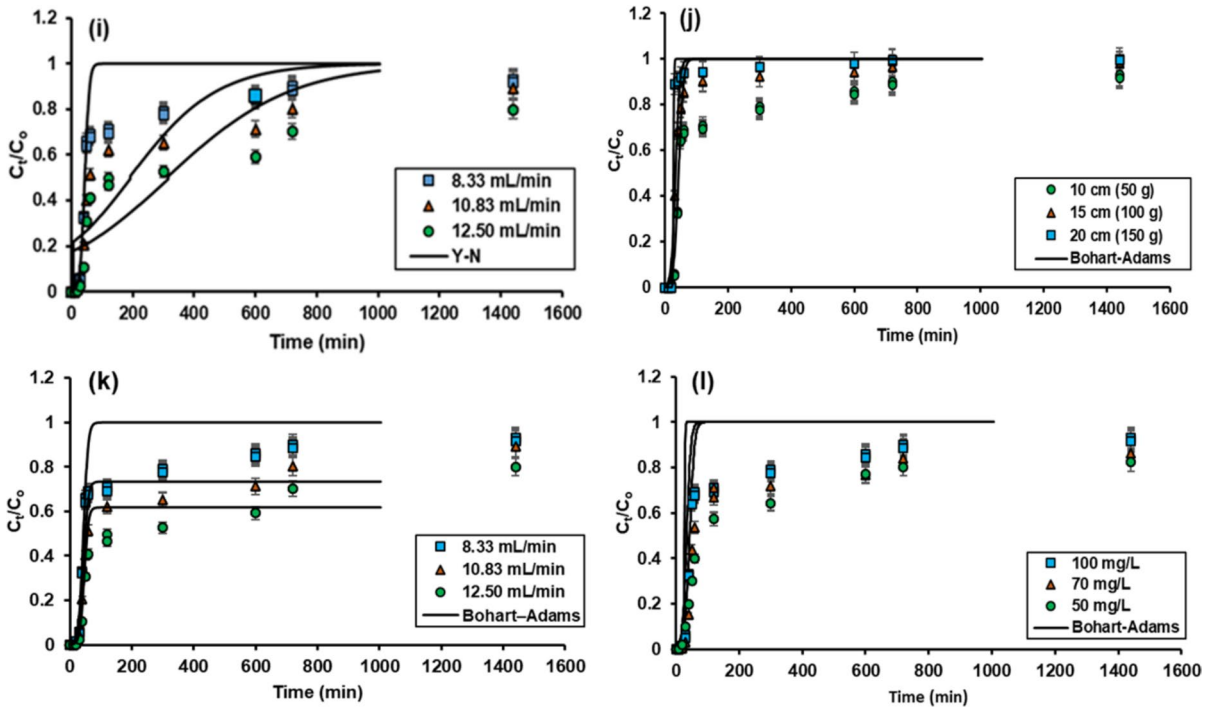


Fig. 8 (continued)

Modelling of column adsorption

In a continuous mode investigation, the findings suggested that greater bed height reduces flow velocity, and higher dye concentration promotes TTZ sorption onto PTC biochar. The results of modeling on column adsorption, Fig. 8 (a-l) and Table 5, show that after increasing bed height from 10 to 20 cm, reducing flow rate from 12.50 to 8.33 mL/min and increasing initial dye concentration from 50 to 100 mg/L, the best-fitted model was Clark model on the based of R^2 values that related to the Freundlich model. The batch adsorption study also concluded that the Freundlich model was best fit for the adsorption of TTZ on PTCs.

According to the Thomas model, the transverse liquid film content across the material's surface equals the forces of intraparticle diffusion (Suter et al. 2025). According to Yoon–Nelson column model, the pace at which the adsorption probability for all adsorbate molecules decreases is proportional to the likelihood that adsorbate molecules will adsorb and that adsorbate will break through the adsorbent (Bakhta et al. 2024). A key component of Bohart–Adams model is the idea that equilibrium is not achievable instantly. The amount of the adsorbate and the residual capacity of the adsorbent determine the rate of adsorption (Mani and Bhandari 2022).

According to the R^2 values, the Clark model fits better under a variety of circumstances than the Y-N(Yoon Nelson), Bohart–Adams (B-A) and Thomas models. When the height of the bed was increased, the Y-N model (0.98), the Thomas model (0.98), Bohart–Adams (0.99) and the Clark model (0.99) fitted almost equally to the column adsorption data. As the initial dye concentration increased, the Thomas, Y-N and B-A models fitted less efficiently (R^2 of 0.85 for each) than the Clark model with R^2 value of 0.97. Likewise, with reducing flow rate, the R^2 value was high for the Clark model (0.92) than other models. As per the results the adsorption efficacy improved by increasing bed height and inlet dye concentration, as indicated by the high R^2 values (0.99, 0.97, and 0.96, respectively). Furthermore, lowering the flow rate improved the bed's adsorption capacity, confirming the Clark model's accuracy in predicting adsorption behavior in these circumstances (Ali et al. 2022b). The breakthrough time of different adsorbents for TTZ removal is shown in Table 6.

Mechanism of adsorption

When determining the adsorption mechanism, several things need to be considered. For instance, the slowest phase in the process must be identified; both the diffusion of compounds inside the adsorbent particles and the transport of substances from the external environment affect the efficiency of adsorption, and how quickly the contaminant moves across a liquid boundary layer or via small pores determines how quickly it clings to a surface overall.

To study how PTCs remove TTZ, we did post-adsorption analyses, such as SEM and FTIR. We also did experiments to look at parameters like the pH of the solution and how

Table 5 Parameters of Clark model, Y-N model, and Thomas model

Clark model			
Parameters	K_c (mL/min mg)	A	R^2
Effect of column height (cm)			
10	0.15	0.82	0.97
15	0.15	0.92	0.98
20	0.32	0.95	0.99
Effect of concentration (mg/L)			
100	0.15	0.82	0.97
70	0.11	0.72	0.97
50	0.06	0.73	0.95
Effect of flow rate (mL/min)			
8.33	0.15	0.82	0.97
10.83	0.10	0.73	0.95
12.50	0.11	0.62	0.92
Y-N model			
parameters	K_{YN} (mL/min mg)	T (min)	R^2
Effect of column height (cm)			
10	0.10	47.8	0.85
15	0.12	35.6	0.96
20	0.85	27.5	0.98
Effect of Concentration (mg/L)			
100	0.10	47.8	0.88
70	0.03	74.7	0.78
50	0.005	255	0.76
Effect of flow rate (mL/min)			
8.33	0.10	47.8	0.88
10.83	0.001	261.9	0.68
12.50	0.003	441.7	0.68
Thomas model			
Parameters	K_T (mL/min mg)	q_{ac} (mg/g)	R^2
Effect of column height (cm)			
10	0.05	7.94	0.86
15	0.82	8.27	0.96
20	18.58	8.32	0.98
Effect of concentration (mg/L)			
100	2.43	7.94	0.88
70	1.81	3.15	0.78
50	6.47	2.61	0.68
Effect of flow rate (mL/min)			
8.33	0.18	7.94	0.97
10.83	0.73	6.63	0.95
12.50	0.61	4.70	0.92
Bohart–Adams model			
Parameters	K_{BA} (mL/min mg)	N_o (mg/L)	R^2
Effect of column height (cm)			
10	0.06	86.98	0.84
15	0.83	5.26	0.95
20	18.59	1.26	0.99

Table 5 (continued)

Bohart–Adams model			
Effect of concentration (mg/L)			
100	0.06	87.08	0.86
70	3.29	3.47	0.78
50	3.29	3.47	0.74
Effect of flow rate (mL/min)			
8.33	0.06	87.10	0.86
10.83	3.28	3.47	0.81
12.50	3.29	3.48	0.54

Table 6 Breakthrough time of different adsorbents during TTZ adsorption in column

Adsorbent	Breakthrough time (min)	Adsorption capacity (mg/g)	Reference
polystyrene/magnetite nanocomposite	150	0.51	(Ali et al. 2022a)
Activated carbon (apricot stones)	Not stated	1.7–2.5	(Albroomi et al. 2016)
activated carbon fiber	43	2.15	(Sarıcı-Özdemir 2012)
Parthenium hysterophorus biochar	150	8.32	This study

long the materials were in contact with each other. The SEM analysis after TTZ adsorption, Fig. 9 (b-d), showed that TTZ stuck to the surface of PTCs. Also, the post-adsorption FTIR results (Fig. 9a) for the TTZ-PTCs mix showed that the peak intensity decreased and the adsorption bands moved. These observations hint that chemisorption might be happening, and this was backed up by studying the speed of the reactions (Perveen et al. 2022).

Electrostatic forces cause the charged areas of the adsorbate to interact with the hydroxyl groups on the PTC surface, as shown in Fig. 10. TTZ removal is also facilitated by Van der Waals forces, hydrogen bonds, and π-π stacking interactions, in addition to the PTCs' hydrophobic alterations and variety of surface functional groups. TTZ adsorption by physisorption could be made possible by PTCs' high surface area and porous nature (Reck et al. 2020). TTZ's anionic nature is influenced by its sulfonate and carboxyl groups, which draw the positively charged PTCs surface at pH 6. The graphitic structure of PTCs is confirmed by FTIR analysis to be enriched in electron-donor groups such as carbonyl and hydroxyl, which interact with the benzene ring of TTZ through π–π interactions (Camara et al. 2020). Furthermore, the carbonyl groups of TTZ and the functional groups of PTCs form a hydrogen bond (Vieira et al. 2018). In Fig. 10 TTZ adsorption process may experience both internal pore diffusion resistance and exterior surface diffusion resistance as a result of the TTZ molecule's huge size. The entire adsorption rate is anticipated to be more significantly influenced by the intraparticle diffusion resistance, though, because big molecules experience steric hindrance and delayed transportation within the PTC's pore structure (Barzallo et al. 2025). We suggest several mechanisms may be involved in the adsorption of TTZ by PTCs, such as electrostatic interaction, pore-filling, surface complexation, π–π interaction, H-bonding, and hydrophobic interaction.

Conclusion

A new carbonaceous adsorbent based on weeds that has an effective TTZ adsorption capability was successfully evaluated in this study. The thermal stability and adsorption performance of PTCs were examined by preparing them at three pyrolysis temperatures (300 °C, 500 °C, and 700 °C). Initial TTZ concentration of 5 mg/L (achieving near-complete removal), pH 6 (maximizing electrostatic attraction between the positively charged PTC surfaces and anionic TTZ), adsorbent dosage of 1.5 g/L (balancing removal efficiency and capacity), and a contact time of 45 min (enough to reach adsorption equilibrium) were the optimized conditions for TTZ removal in batch experiments. PTC 500 demonstrated better removal (98.9%) than PTC 300 (89.7%) and PTC 700 (95.4%), presumably as a result of the ideal pore structure and availability of active sites at intermediate temperatures. Both batch and fixed-bed column adsorption demonstrated strong adsorption performance for PTC 500. In addition to the various chemical groups on its surface, forces like Van der Waals forces, hydrogen bonds, and π–π stacking interactions played a significant role in removing TTZ. According to the findings of the batch adsorption trials, TTZ's C-π might be used with PTC 500. The maximum adsorption capacity determined by the Langmuir model was 95.75 mg/g. The isotherm and kinetic data were well-fitted to the Langmuir and pseudo-first-order models, respectively. The outcomes of fixed-bed column adsorption tests clearly showed the potential of the adsorbent in industrial

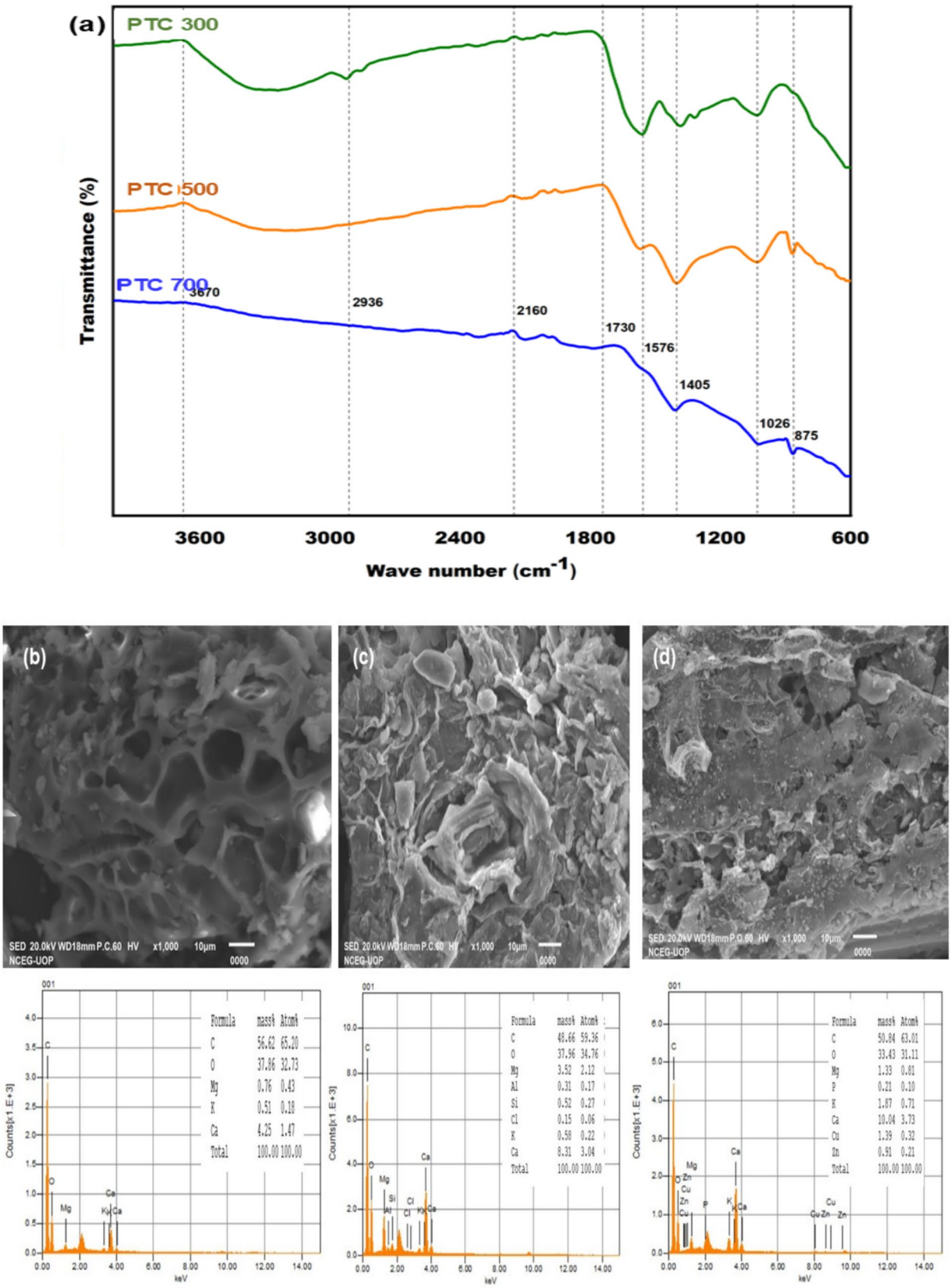


Fig. 9 **a** FTIR, and **b**–**d** SEM images with EDX spectra of (PTC 300, PTC 500 and PTC 700), respectively, after adsorption of TTZ from wastewater

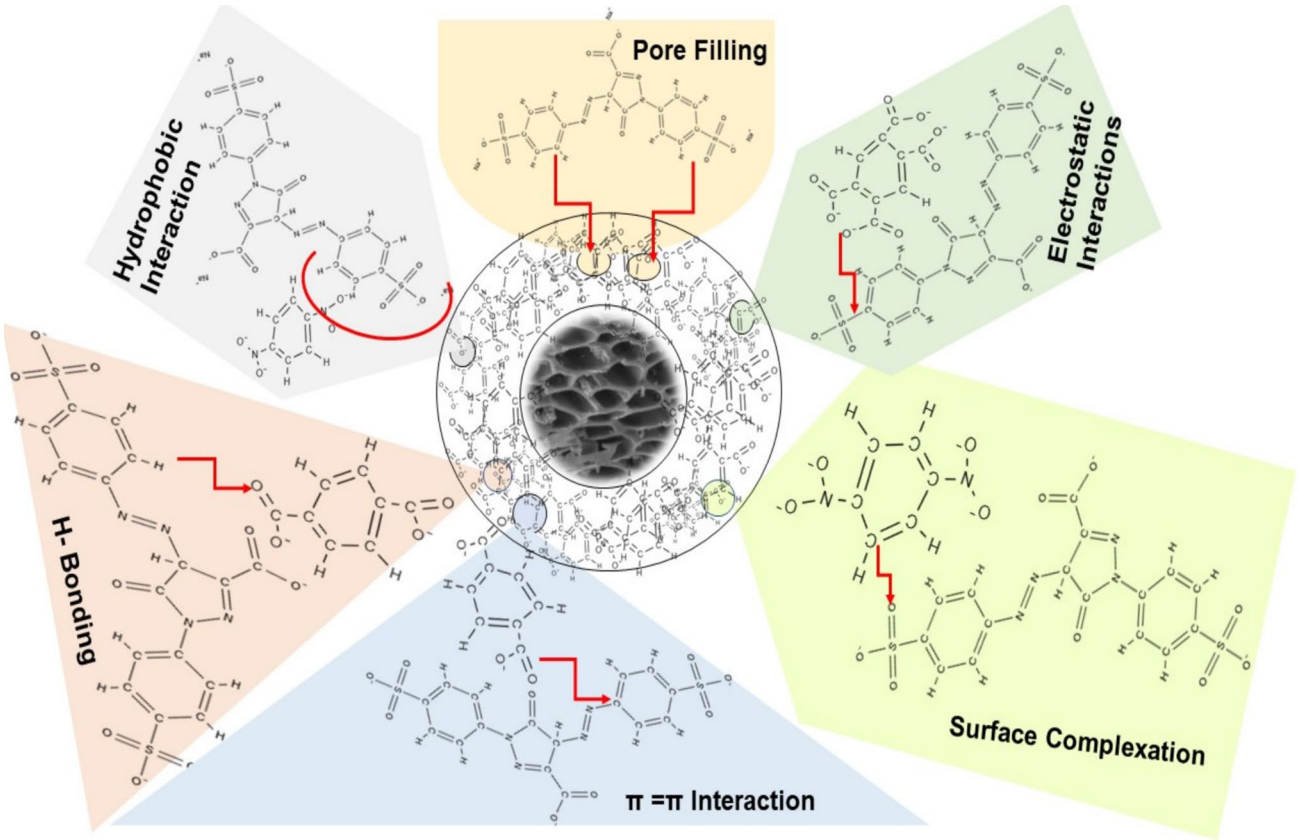


Fig. 10 Overall mechanism of adsorption of TTZ dye

applications. Even in solutions with large quantities of coexisting ions, TTZ could be effectively removed. Results indicate that increased TTZ sorption onto PTC is promoted by higher initial dye concentration, lower flow velocity, and higher bed height; the Clark model is the best match. TTZ's comparatively large molecular size means that both intraparticle diffusion within the biochar's pores and exterior film diffusion via the outer layer affect adsorption. The batch tests, with an equilibrium uptake of 95.57 mg/g, indicate that intraparticle diffusion is important, as the dye molecules must overcome pore resistance to reach interior locations. This is in line with the assumptions of the Langmuir and pseudo-first-order models. However, the highest adsorption capacity in the fixed-bed column was only 8.32 mg/g, suggesting that fluctuating conditions have a greater impact on external mass transfer over the outer layer. The significance of film diffusion in explaining breakthrough behavior was further supported by the Clark model, which offered the greatest match to column data. As a result, even if both resistances are active, film diffusion has a stronger influence during column operation, while pore diffusion primarily regulates the batch adsorption process. At least five adsorption–desorption cycles could be performed on the saturated columns by regenerating them with 0.1 N NaOH. As a result, PTC 500 may prove to be a useful adsorbent for the removal of TTZ in future applications. Future studies should focus on enhancing PTC preparation, evaluating its effectiveness in various scenarios, and assessing its long-term stability. A better comprehension of the sorption mechanisms can be obtained by gaining mechanistic insights into the interactions between PTCs and TTZ. Evaluating PTCs' environmental impact and economic viability, as well as their suitability for large-scale applications and real-world situations, will be essential. PTCs' usefulness can be enhanced by investigating their potential for eliminating additional pollutants and exploring their various applications.

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Data availability Data would be made available at a reasonable request.

Declarations

Conflict of interest Authors declare that they have no known financial or non-financial competing interests that are directly or indirectly related to the work submitted for publication.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

References

Ahmed HR, Salih MA, Agha NN, Tofiq DI, Karim MA, Kayani KF, Ealias AM, Hamarawf RF, Aziz BK, Khatab LO (2024) Enhanced methyl orange removal in aqueous solutions using bio-catalytic metal oxides derived from pomegranate peel waste: a green chemistry evaluation. React Kinet Mech Catal 137(6):3417–3440

Albadarin AB, Charara M, Tarboush B, Ahmad M, Kurniawan TA, Naushad M, Walker GM, Mangwandi C (2017) Mechanism analysis of tartrazine biosorption onto masau stones; a low cost by-product from semi-arid regions. J Mol Liq 242:478–483

Albroomi HI, Elsayed MA, Baraka A, Abdelmaged MA (2016) Batch and fixed-bed adsorption of tartrazine azo-dye onto activated carbon prepared from apricot stones. Appl Water Sci 7(4):2063–2074. https://doi.org/10.1007/s13201-016-0387-2

Alegbe EO, Uthman TO (2024) A review of history, properties, classification, applications and challenges of natural and synthetic dyes. Heliyon 10:e33646

Ali MA, Mubarak MF, Keshawy M, Zayed MA, Ataalla M (2022a) Adsorption of tartrazine anionic dye by novel fixed bed Core-Shell- polystyrene Divinylbenzene/Magnetite nanocomposite. Alexandria Eng J 61:1335–1352

Ali MA, Mubarak MF, Keshawy M, Zayed MA, Ataalla M (2022b) Adsorption of tartrazine anionic dye by novel fixed bed Core-Shell-polystyrene Divinylbenzene/Magnetite nanocomposite. Alexandria Eng J 61:1335–1352

Al-Mahbashi NMY, Kutty SRM, Bilad MR, Huda N, Kobun R, Noor A, Jagaba AH, Al-Nini A, Ghaleb AAS, Al-Dhawi BNS (2022) Bench-scale fixed-bed column study for the removal of dye-contaminated effluent using sewage-sludge-based biochar. Sustainability 14:6484

Almeida END, Pacheco EA, Nagahama KJ, Santos TSM, Souza AOD (2024) Adsorption kinetics of tartrazine dye on activated carbon derived from guava (Psidium guajava) seeds. J Brazilian Chem Soc. 35:e-20240139

Almeida-Naranjo CE, Frutos M, Guerrero VH, Villamar-Ayala C (2024) Characterization and performance of peanut shells in caffeine and triclosan removal in batch and fixed-bed column tests. Molecules 29(12):2923. https://doi.org/10.3390/molecules29122923

Amaku JF, Oyedotun KO, Maxakato NW, Akpomie KG, Okeke ES, Olisah C, Malloum A, Adegoke KA, Ighalo JO, Conradie J, Ohoro CR (2024) Aqueous phase removal of tartrazine. Chemistry Africa, Springer science and business media Deutschland GmbH

Ambaye TG, Vaccari M, VAN Hullebusch ED, Amrane A, Rtimi S (2020) Mechanisms and adsorption capacities of biochar for the removal of organic and inorganic pollutants from industrial wastewater. Int J Environ Sci Technol 18(10):3273–3294. https://doi.org/10.1007/s13762-020-03060-w

APHA 2017. *Standard methods for the examination of water and wastewater*, American Public Health Association

Bakhta S, Sadaoui Z, Bouazizi N, Samir B, Cosme J, Allalou O, le Derf F, VIEILLARD, (2024) Successful removal of fluoride from aqueous environment using Al (OH) 3@ AC: column studies and breakthrough curve modeling. RSC Adv 14:1–14

Banerjee S, Chattopadhyaya MC (2017) Adsorption characteristics for the removal of a toxic dye, tartrazine from aqueous solutions by a low cost agricultural by-product. Arab J Chem 10:S1629–S1638

Bankole DT, Inyinbor AA, Oluyori AP, Arowolo MO (2024) Adsorptive removal of synthetic food dyes using low-cost biochar: efficiency prediction, kinetics and desorption index evaluation. Bioresour Technol Rep 25:101709

Barciela P, Perez-Vazquez A, Prieto MJF (2023) Azo dyes in the food industry: features, classification, toxicity, alternatives, and regulation. Food Chem Toxicol 178:113935

Barzallo D, Carrasquero E, Andrade M, HEREDIA DA, JARA, PALMAY P (2025) Preparation and characterization of unactivated, activated, and γ-Fe2o3 nanoparticle-functionalized biochar from rice husk via pyrolysis for dyes removal in aqueous samples: comparison, performance, and mechanism. ChemEng 9(2):30

Camara AS, Lütke SF, Pinheiro CP, Vieira MLG, JR SANT'ANNA CADAVAL TR, DE ALMEIDA PINTO LA (2020) Chitosan-coated sand and its application in a fixed-bed column to remove dyes in simple, binary, and real systems. Environ Sci Pollut Res Int 27:37938–37945

Chukwuemeka-Okorie HO, Ekuma FK, Akpomie KG, Nnaji JC, Okereafor AG (2021) Adsorption of tartrazine and sunset yellow anionic dyes onto activated carbon derived from cassava sievate biomass. Appl Water Sci 11:1–8

Ebeid H, di Gianvincenzo F, KRALJ CIGIĆ, I. & STRLIČ, M. (2024) Chromatographic analysis of natural dyes in mediaeval Islamic paper. Herit Sci 12:13

Fernandez RMD, Estrada RJR, Tomon TRB, JR DINGCONG RG, JR AMPARADO RF, CAPANGPANGAN RY, MALALUAN RM, DUMANCAS GG, LUBGUBAN AA, ALGUNO AC (2023) Experimental design and breakthrough curve modeling of fixed-bed columns utilizing a novel 3D coconut-based polyurethane-activated carbon composite adsorbent for lead sequestration. Sustainability 15:14344

Hasan M, Al Biruni MT, Azad S, Ahmed T, BIOREFINERY (2024) Adsorptive removal of dye from textile wastewater employing *Moringa oleifera* leaves biochar as a natural biosorbent. Biomass Convers Biorefin 14:11075–11091

Irfan I, Inam MA, Usmani W, Iftikhar R, Jahan Z (2023) Adsorptive recovery of phosphate using iron functionalized biochar prepared via co-pyrolysis of wheat straw and sewage sludge. Environ Technol Innov. https://doi.org/10.1016/j.eti.2023.103434

Jamil MD, Waheed M, Akhtar S, Bangash N, Chaudhari SK, Majeed M, Hussain M, Ali K, Jones DA (2022) Invasive plants diversity, ecological status, and distribution pattern in relation to edaphic factors in different habitat types of district Mandi Bahauddin, Punjab, Pakistan. Sustainability 14(20):13312. https://doi.org/10.3390/su142013312

Khalid U, Inam MA (2024) The influence of pyrolysis temperature on the performance of cotton stalk biochar for hexavalent chromium removal from wastewater. Water Air Soil Pollut. https://doi.org/10.1007/s11270-024-06922-y

Kundu D, Sharma P, Bhattacharya S, Gupta K, Sengupta S, Shang J (2024) Study of methylene blue dye removal using biochar derived from leaf and stem of *Lantana camara* L. Carb Res 3:22

Mahmoud ME, Abdelfattah AM, Tharwat RM, Nabil GMJJOML (2020) Adsorption of negatively charged food tartrazine and sunset yellow dyes onto positively charged triethylenetetramine biochar: Optimization, kinetics and thermodynamic study. J Mol Liq 318:114297

Malik SM, Nawaz MS, Ali M, Manzoor K, Khan SJ, Ghaffour N (2021) Treatment of high-strength synthetic textile wastewater through anaerobic osmotic membrane bioreactor and effect of sludge characteristics on flux. Environ Qual Manag 31:85–98

Mani SK, Bhandari R (2022) Efficient fluoride removal by a fixed-bed column of self-assembled Zr(IV)-, Fe(III)-, Cu(II)-complexed polyvinyl alcohol hydrogel beads. ACS Omega 7:15048–15063

Mondal S, Aikat K, Siddharth K, Sarkar K, Daschaudhury R, Mandal G, Halder G (2017) Optimizing ranitidine hydrochloride uptake of *Parthenium hysterophorus* derived N-biochar through response surface methodology and artificial neural network. Process Saf Environ Prot 107:388–401

Nawaz MS, Gadelha G, Khan SJ, Hankins N (2013) Microbial toxicity effects of reverse transported draw solute in the forward osmosis membrane bioreactor (FO-MBR). J Membr Sci 429:323–329

Okeola F, Odebunmi E, Ameen O, Amoloye M, Lawal A, Abdulmummeen A (2017) Equilibrium kinetics and thermodynamic studies of the adsorption of tartrazine and sunset yellow. Arid Zone J Eng Technol Environ 13:268–280

Pandey D, Daverey A, Dutta K, Arunachalam KJ (2023) Dye removal from simulated and real textile effluent using laccase immobilized on pine needle biochar. O w p e 53:103710

Perveen S, Nadeem R, Nosheen F, Tongxiang L, Anwar T (2022) Synthesis of biochar-supported zinc oxide and graphene oxide/zinc oxide nanocomposites to remediate tartrazine dye from aqueous solution using fixed-bed column reactor. Appl Nanosci 12:1491–1505

Pirarath R, Aljafari B, Wu JJ, Anandan S (2023) Removal of tartrazine dye and mercury present in aqueous solutions using hexamethylenetetramine exfoliated MoS_2 nanosheets as adsorbent: a comparison of kinetic and isotherm models. Bull Mater Sci 46:67

Qurat Ul A, Shafiq M, Capareda SC, Firdaus E B (2021) Effect of different temperatures on the properties of pyrolysis products of *Parthenium hysterophorus*. J Saudi Chem Soc 25:101197–101197

Rambhatla N, Panicker TF, Mishra RK, Manjeshwar SK, Sharma A (2025) Biomass pyrolysis for biochar production: study of kinetics parameters and effect of temperature on biochar yield and its physicochemical properties. Results Eng 25:103679

Ravindiran G, Sundaram H, Rajendran EM, Ramasamy S, Nabil A-Z, Ahmed B (2023) Removal of azo dyes from synthetic wastewater using biochar derived from sewage sludge to prevent groundwater contamination. Urban Clim 49:101502

Rawat S, Samreen K, Nayak AK, Singh J, Koduru JR (2021) Fabrication of iron nanoparticles using *Parthenium*: a combinatorial eco-innovative approach to eradicate crystal violet dye and phosphate from the aqueous environment. Environ Nanotechnol Monit Manag 15:100426

Reck IM, Paixão RM, Bergamasco R, Vieira MF, Vieira AMSJSS, TECHNOLOGY (2020) Investigation of Moringa oleifera seeds as effective and low-cost adsorbent to remove yellow dye tartrazine in fixed-bed column. Sep Sci Technol 55:13–25

Rzig B, Kojok R, Khalifa EB, Magnacca G, Lahssini T, Hamrouni B, Bellakhal N (2024) Adsorption performance of tartrazine dye from wastewater by raw and modified biomaterial: equilibrium, isotherms, kinetics and regeneration studies. Biomass Convers Biorefin 14:18313–18330

Sadaf S, Bhatti HN (2014) Batch and fixed bed column studies for the removal of Indosol Yellow BG dye by peanut husk. J Taiwan Inst Chem Eng 45:541–553

SARıCı-ÖZDEMIR Ç (2012) Modeling of tartrazine adsorption onto activated carbon fiber in a continuous fixed-bed reactor. Desalin Water Treat 46:234–243

Shams M, Nezhad NT, Dehghan A, Paydar M, Mohammadi AA, Zarei A (2020) Heavy metals exposure , carcinogenic and non- carcinogenic human health risks assessment of groundwater around mines in Joghatai , Iran. Int J Environ Anal Chem 00:1–16

Singh K, Azad SK, Dave H, Prasad B, Maurya DM, Kumari M, Dubey D, Rai A, Sillanpää M, Sah MPJBC, BIOREFINERY, (2024) Effective removal of Cr (VI) ions from the aqueous solution by agro-waste-based biochar: an exploration of batch and column studies. Biomass Convers Biorefin 14:19215–19229

Song Q, Kong F, Liu B-F, Song X, Ren H-Y (2024) Biochar-based composites for removing chlorinated organic pollutants: applications, mechanisms, and perspectives. Environ Sci Ecotechnol 21:100420

Soran M-L, Bocșa M, Pintea S, Stegarescu A, Lung I, Opriş O (2023) Commercially biochar applied for tartrazine removal from aqueous solutions. A S 14:53

Suter E, Rutto H, Omwoyo W (2025) Continuous packed bed column removal of Cr6+, Cd2+, and Pb2+ ions from synthetic wastewater using polymeric ultra-permeable and biodegradable ferromagnetic nanocomposite membrane. Can J Chem Eng 103:2642–2658

Vieira ML, Martinez MS, Santos GB, Dotto GL, Pinto LA (2018) Azo dyes adsorption in fixed bed column packed with different deacetylation degrees chitosan coated glass beads. j Environ Chem Eng 6:3233–3241

Yagub MT, Sen TK, Afroze S, Ang HM (2014) Dye and its removal from aqueous solution by adsorption: a review. Adv Colloid Interface Sci 209:172–184. https://doi.org/10.1016/j.cis.2014.04.002

Yan J, Xue Y, Long L, Zeng Y, Hu X (2018) Adsorptive removal of As(V) by crawfish shell biochar: batch and column tests. Environ Sci Pollut Res Int 25:34674–34683

Yasmeen M, Nawaz MS, Manzoor K, Khan SJ, Ghaffour N (2023) Evaluating dye recovery and reusability potential from dyebath effluent using forward osmosis membranes for minimum liquid discharge. Chemosphere 338:139433–139433

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